Exam 02 Review — Modules 7–11

BIOL-1

Name: _____ **Date:** _____

Learning Objectives

By completing this review, you will:

1. Reinforce your understanding of the Central Dogma (DNA → RNA → Protein).
2. Compare mitosis and meiosis and their roles in the body.
3. Solve genetics problems using Punnett squares.
4. Explain how gene expression is regulated and what epigenetics means.
5. Describe the tools of biotechnology (PCR, gel electrophoresis, CRISPR).

Overview

This lab is your comprehensive review of **Modules 7–11** in preparation for Exam 02. It covers the major concepts from Molecular Genetics, Cellular Genetics, Inheritance Genetics, Epigenetics, and Genomics & Biotechnology. Use this as a self-assessment: if you can answer these questions confidently, you are ready for the exam.

Materials

- This worksheet
- Codon table (from Lab 7 or your Key Points)
- Scratch paper for Punnett squares

Part 1: Molecular Genetics (Module 7)

1. Fill in the Central Dogma:

DNA → ___ → ___

2. What are the base pairing rules for DNA? For RNA?

3. Name the enzyme responsible for each process:

Process	Enzyme/Structure
DNA Replication	
Transcription	
Translation	

4. Given the DNA template strand: 3′-TACGCAATT-5′, write the mRNA sequence and use a codon table to determine the amino acids.

- mRNA:
- Amino acids:

5. What is the difference between a point mutation and a frameshift mutation? Which is usually more harmful and why?

Part 2: Cellular Genetics (Module 8)

6. Fill in the comparison table:

Feature	Mitosis	Meiosis
Number of divisions		
Number of daughter cells		
Ploidy of daughter cells		
Genetically identical?		
Purpose		

7. What separates during Anaphase I of meiosis? What separates during Anaphase II?

8. Name the three sources of genetic variation in sexual reproduction:

1.
2.
3.

9. What is nondisjunction? Name one condition it can cause.

10. What usually happens when the cell cycle loses its normal checkpoints and divides uncontrollably?

Part 3: Inheritance Genetics (Module 9)

11. Cross two heterozygous tall plants (Tt × Tt). Fill in the Punnett square:

	T	t
T		
t		

- Phenotypic ratio:

12. In snapdragons, Red (RR) × White (R′R′) = Pink (RR′). Cross two pink flowers (RR′ × RR′). What is the phenotypic ratio?

13. A mother is Type A (I^Ai) and a father is Type B (I^Bi). Can they have a Type O child? Show the cross.

14. A carrier mother (X^B X^b) for color blindness and a normal father (X^B Y) have children. What fraction of sons will be colorblind?

15. Two unaffected parents have an affected child. Is the trait most likely dominant or recessive? Explain.

Part 4: Epigenetics (Module 10)

16. All cells have the same DNA. How do muscle cells and brain cells become different?

17. Describe how the lac operon works. What happens when lactose is absent? When it is present?

18. Fill in the table:

Mechanism	Effect on Gene Expression
DNA methylation	
Histone acetylation	

19. How do calico cats demonstrate X-inactivation?

20. What is the key difference between an epigenetic change and a genetic mutation?

Part 5: Genomics & Biotechnology (Module 11)

21. What do restriction enzymes do? What does DNA ligase do?

22. Describe the three steps of one PCR cycle:

Step	What Happens
Denaturation	
Annealing	
Extension	

23. In gel electrophoresis, do smaller or larger DNA fragments travel farther? Why?

24. What is CRISPR-Cas9? How is it different from restriction enzymes?

25. What is a genetically modified organism (GMO)? Give one example from agriculture or medicine.

Part 6: Connections Across Modules

26. Trace the path from a gene to a trait. Start with DNA in the nucleus and end with a visible phenotype. Use these terms: transcription, mRNA, translation, protein, phenotype.

27. How does Mendel's Law of Segregation (Module 9) connect to the physical process of homolog separation during Meiosis I (Module 8)?

28. A tumor suppressor gene is silenced by DNA methylation (no mutation in the DNA). Could this lead to cancer? Explain using concepts from both Module 8 (cancer) and Module 10 (epigenetics).

29. Scientists insert a human insulin gene into a bacterial plasmid. The bacteria successfully produce human insulin. Why does this work? (Connect the universal genetic code from Module 7 to the biotechnology tools from Module 11.)

30. In your own words, explain the most important concept you learned across Modules 7–11 and why it matters for understanding biology.